Technical Report - FastSpeech2 TTS

1. Implementation: ming024 FastSpeech2, variance adaptor with duration/pitch/energy predic
tors, encoder 4 layers, decoder 4 (English) /6 (Telugu).

2. Dataset Selection: Svarah filtered 1-10 sec, random subsample 5h train 1h val. IndicTTS
Telugu similar, no overlap, seed 42.

3. Preprocessing: English g2p-en, Telugu Unicode NFC, pyworld for F0, librosa for mel.

4. Training: English lr 1e-4 batch16 1000 epochs, Telugu lr 8e-5 batch12 1200 epochs.

5. Loss curves: Converge after warmup 4000 steps, Telugu slower.

6. Speech Quality: Handles antidisestablishmentarianism, homographs conduct, rhythm, time-
date 10:45 AM.

7. Telugu: Handles long compounds, code-mixed andaroo via transliteration.

8. Variance plots: duration correlates syllable length, pitch rises for emphasis, energy f
or stress.

9. Comparative: English better for numbers, Telugu needs more data for prosody.

10. Reproducibility: pip install -r requirements.txt, python preprocess.py train.py infere
nce.py